

# COMBINATORIAL CATEGORICAL EQUIVALENCES; POST ARXIV POSTING

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## 1. INTRODUCTION

Here are some notes on thoughts since submitting [arXiv:1402.7151](https://arxiv.org/abs/1402.7151). They concern a general setting with a view to including the example of  $\Omega_n$  in the sense of [2].

The intention of this paper is to prove a class of equivalences of categories that seem of interest in enumerative combinatorics as per [19], representation theory as per [10], and homotopy theory as per [3]. More specifically, for a class of categories  $\mathcal{P}$ , we construct a category  $\mathcal{D}$  with zero morphisms (that is,  $\mathcal{D}$  has homs enriched in the category  $1/\text{Set}$  of pointed sets) and an equivalence of categories of the form

$$[\mathcal{P}, \mathcal{X}] \simeq [\mathcal{D}, \mathcal{X}]_{\text{pt}} . \quad (1.1)$$

On the left-hand side we have the usual category of functors from  $\mathcal{P}$  into any additive category  $\mathcal{X}$  which has finite direct sums and splitting for idempotents. On the right-hand side we have the category of functors which preserve the zero morphisms.

One example has  $\mathcal{P} = \Delta_{\perp, \top}$ , the category whose objects are finite ordinals with first and last element, and whose morphisms are functions preserving order and first and last elements. Then  $\mathcal{D}$  is the category with non-zero and non-identity morphisms

$$0 \xleftarrow{\partial} 1 \xleftarrow{\partial} 2 \xleftarrow{\partial} \dots$$

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such that  $\partial \circ \partial = 0$ . Since there is an isomorphism of categories

$$\Delta_{\perp, \top} \cong \Delta_+^{\text{op}}, \quad (1.2)$$

where the right-hand side is the algebraist's simplicial category (finite ordinals and all order-preserving functions), our result (1.1) reproduces the Dold-Puppe-Kan Theorem [12, 13, 22].

Cubical sets also provide an example of our setting; see Example 4.3. We conclude that cubical simplicial abelian groups are equivalent to semi-simplicial abelian groups.

For a category  $\mathcal{A}$  equipped with a suitable factorization system  $(\mathcal{E}, \mathcal{M})$  [14], write  $\text{Par}\mathcal{A}$  (strictly the notation should also show the dependence on  $\mathcal{M}$ ) for the category with the same objects as  $\mathcal{A}$  and with  $\mathcal{M}$ -partial maps as morphisms. We identify  $\mathcal{E}$  with the subcategory of  $\mathcal{A}$  having the same objects but only the morphisms in  $\mathcal{E}$ . Assume each object of  $\mathcal{A}$  has only finitely many  $\mathcal{M}$ -subobjects. Let  $\mathcal{X}$  be any additive category with finite direct sums and splitting idempotents. Our main equivalence (1.1) includes as a special case an equivalence of categories

$$[\text{Par}\mathcal{A}, \mathcal{X}] \simeq [\mathcal{E}, \mathcal{X}]. \quad (1.3)$$

(Here  $\mathcal{D}$  is obtained from  $\mathcal{E}$  by freely adjoining zero morphisms.) Indeed, we show that this special case implies the general case.

Let  $\mathfrak{S}$  be the groupoid of finite sets and bijective functions. Let  $\text{FI}\sharp$  denote the category of finite sets and injective partial functions. Let  $\text{Mod}^R$  denote the category of left modules over the ring  $R$ . Our original motivation was to understand and generalize the classification theorem for  $\text{FI}\sharp$ -modules appearing as Theorem 2.24 of [10], which provides an equivalence

$$[\text{FI}\sharp, \text{Mod}^R] \simeq [\mathfrak{S}, \text{Mod}^R]$$

between the category of functors  $\text{FI}\sharp \rightarrow \text{Mod}^R$  and the category of functors  $\mathfrak{S} \rightarrow \text{Mod}^R$ . This is the special case of (1.3) above in which  $\mathcal{A}$  is the category  $\text{FI}$  of finite sets and injective functions, and  $\mathcal{M}$  consists of all the morphisms. This result has provided a new viewpoint on representations of the symmetric groups, and a new viewpoint on Joyal species [19, 20].

In order to consider stability properties of representations of the symmetric groups, the authors of [10] also consider  $\text{FI}$ -modules: that is,  $R$ -module-valued functors from the category  $\text{FI}$ . Each  $\text{FI}\sharp$ -module clearly has an underlying  $\text{FI}$ -module, so their Theorem 2.24 shows how symmetric group representations become  $\text{FI}$ -modules. One application they give is a structural version of the Murnaghan Theorem [29, 30], a problem which has its combinatorial aspects [35]. We are reminded of the way in which Mackey functors [26, 31] give extra freedom to representation theory.

Another instance of an equivalence of the form (1.3) is when  $\mathcal{A}$  is the category of finite sets with its usual (surjective, injective)-factorization system. Then  $\text{Par}\mathcal{A}$  is equivalent to the category of pointed finite sets, which is equivalent to Graeme Segal's category  $\Gamma$  [33]. After completing this work, we were alerted to Teimuraz Pirashvili's interesting paper [32] which gives this finite sets example, makes the connection with Dold-Puppe-Kan, and discusses stable homotopy of  $\Gamma$ -spaces.

Consider the basic equivalence (1.1). It is really about Cauchy (or Morita) equivalence of the free additive category on the ordinary category  $\mathcal{P}$  and the free additive category on the category with zero morphisms  $\mathcal{D}$ . By the general theory of Cauchy

completeness (see [36] for example), to have (1.1) for all Cauchy complete additive categories  $\mathcal{X}$ , it suffices to have it when  $\mathcal{X}$  is the category of abelian groups. Cauchy completeness amounts to existence of absolute limits (see [37]) and, for additive categories, amounts to the existence of finite direct sums and splittings for idempotents.

The strategy of proof of our main equivalence (1.1) is as follows. In Section 2 we provide general criteria for deducing a  $\mathcal{V}$ -functor between  $\mathcal{V}$ -functor  $\mathcal{V}$ -categories is an equivalence given that some other  $\mathcal{V}$ -functor between “smaller”  $\mathcal{V}$ -functor  $\mathcal{V}$ -categories is known to be an equivalence. In Section 3 we describe the setting for our main result and then, in Section 4 show that the partial map, Dold-Puppe-Kan and cube examples satisfy our assumptions. We show in Section 5 that the setting of Section 3 leads to an adjunction between functor categories of a particularly combinatorial kind. Section 6 reduces the proof that the adjunction is an equivalence to the particular case of partial maps, then Section 7 treats the partial map case.

## 2. A GENERAL RESULT

Let  $\mathcal{X}$  be a bicategory. A morphism  $f: A \rightarrow B$  is called a *map* when it has a right adjoint  $f^*: B \rightarrow A$ . The map is called *Cauchy dense* when  $f^*: B \rightarrow A$  is monadic.

Consider the following diagram of two squares, each commuting up to isomorphism.

$$\begin{array}{ccccc} B & \xrightarrow{k^*} & D & \xrightarrow{k} & B \\ m \downarrow & \xleftarrow{\cong} & n \downarrow & \xrightarrow{\cong} & m \downarrow \\ A & \xrightarrow{h^*} & C & \xrightarrow{h} & A \end{array} \quad (2.4)$$

**Proposition 2.1.** *In the diagram (2.4), suppose  $h$  and  $k$  are Cauchy dense maps, and that the two 2-cells are mates under adjunction in the sense of [23]. If  $n$  is an equivalence then so is  $m$ .*

*Proof.* By applying the pseudofunctor  $\mathcal{X}(X, -)$ , the Proposition reduces to a result in Cat. Take a monad  $S$  on a category  $\mathcal{C}$ , a monad  $T$  on a category  $\mathcal{D}$ , and a natural isomorphism

$$\begin{array}{ccc} \mathcal{D}^T & \xrightarrow{M} & \mathcal{C}^S \\ U \downarrow & \xrightarrow{\theta} & \downarrow V \\ \mathcal{D} & \xrightarrow{N} & \mathcal{C} \end{array},$$

where  $\mathcal{C}^S$  is the category of Eilenberg-Moore algebras for  $S$ , similarly  $\mathcal{D}^T$  for  $T$ , and  $V$  and  $U$  are the underlying functors. Let  $G \dashv V$  and  $F \dashv U$  be the usual canonical adjoints. Let  $\phi: GN \Rightarrow MF$  be the mate of  $\theta$  under adjunction. We must show that  $N$  an equivalence and  $\phi$  invertible imply  $M$  an equivalence. It is readily checked that the natural transformation

$$\psi = (SN = VGN \xrightarrow{V\phi} VMF \xrightarrow{\theta^{-1}F} NUF = NT)$$

defines a monad morphism

$$\begin{array}{ccc} \mathcal{D} & \xrightarrow{N} & \mathcal{C} \\ T \downarrow & \xleftarrow{\psi} & \downarrow S \\ \mathcal{D} & \xrightarrow{N} & \mathcal{C} \end{array},$$

inducing a functor  $N^\psi: \mathcal{D}^T \rightarrow \mathcal{C}^S$  isomorphic to  $M$ . If  $\phi$  is invertible, so too is  $\psi$ . So then, if  $N$  is an equivalence, the morphism  $(N, \psi): (\mathcal{D}, T) \rightarrow (\mathcal{C}, S)$  is an equivalence in the 2-category  $\text{Mnd}(\text{Cat}) = \text{Lax}(1, \text{Cat})$  of monads in  $\mathcal{X}$ . Since the Eilenberg-Moore construction  $(-)^{(-)}: \text{Mnd}(\text{Cat}) \rightarrow \text{Cat}$  is a 2-functor, it follows that  $N^\psi$  is an equivalence and so  $M$  is.  $\square$

Let  $\mathcal{V}$  be a symmetric monoidal closed category which is complete and cocomplete. Suppose  $M: \mathcal{B} \rightarrow \mathcal{A}$  is a  $\mathcal{V}$ -module viewed as a  $\mathcal{V}$ -functor  $M: \mathcal{A}^{\text{op}} \otimes \mathcal{B} \rightarrow \mathcal{V}$ . We call  $M$  the *kernel* of the adjunction

$$[\mathcal{A}, \mathcal{V}] \xrightleftharpoons[\widetilde{M}]{\widehat{M}} [\mathcal{B}, \mathcal{V}], \quad (2.5)$$

where

$$\widetilde{M}(T)A = \int_B \mathcal{V}(M(A, B), TB) = [\mathcal{B}, \mathcal{V}](M(A, -), T) \quad (2.6)$$

and

$$\widehat{M}(F)B = \int^B M(A, B) \otimes FA. \quad (2.7)$$

To say  $M: \mathcal{B} \rightarrow \mathcal{A}$  is an equivalence as a  $\mathcal{V}$ -module is the same as saying  $\widetilde{M}: [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$  is an equivalence of  $\mathcal{V}$ -categories. We also say  $\mathcal{A}$  and  $\mathcal{B}$  are *Cauchy* (or *Morita*) equivalent when such an equivalence exists. See [36] for more detail.

A  $\mathcal{V}$ -functor  $H: \mathcal{C} \rightarrow \mathcal{A}$  is Cauchy dense (as a morphism  $H_* = \mathcal{A}(1_{\mathcal{A}}, H): \mathcal{C} \rightarrow \mathcal{A}$  in the bicategory  $\mathcal{V}\text{-Mod}$ ) when the canonical

$$\int^C \mathcal{A}(HC, A_1) \otimes \mathcal{A}(A, HC) \rightarrow \mathcal{A}(A, A_1)$$

is a strong epimorphism in  $\mathcal{V}$  for all  $A, A_1 \in \mathcal{A}$ . When  $\mathcal{V}$  is the category  $\text{Agp}$  of abelian groups this means that each  $A \in \mathcal{A}$  is a retract of a finite direct sum of objects  $HC$  in the image of  $H$ .

Suppose  $M: \mathcal{B} \rightarrow \mathcal{A}$  and  $N: \mathcal{D} \rightarrow \mathcal{C}$  are  $\mathcal{V}$ -modules, and  $H: \mathcal{C} \rightarrow \mathcal{A}$  and  $K: \mathcal{D} \rightarrow \mathcal{B}$  are  $\mathcal{V}$ -functors. Each  $\mathcal{V}$ -natural family

$$\theta_{C,D}: N(C, D) \rightarrow M(HC, KD)$$

induces  $\mathcal{V}$ -natural families  $\theta_{A,D}^r$ :

$$\int^C N(C, D) \otimes \mathcal{A}(A, HC) \xrightarrow{\int^C \theta \otimes 1} \int^C M(HC, KD) \otimes \mathcal{A}(A, HC) \xrightarrow{\text{act}^r} M(A, KD)$$

and  $\theta_{C,B}^\ell$ :

$$\int^D \mathcal{B}(KD, B) \otimes N(C, D) \xrightarrow{\int^D 1 \otimes \theta} \int^D \mathcal{B}(KD, B) \otimes M(C, D) \xrightarrow{\text{act}^\ell} M(HC, B) .$$

**Corollary 2.2.** *Suppose  $H$  and  $K$  are Cauchy dense and both  $\theta^\ell$  and  $\theta^r$  are invertible. If  $\tilde{N}: [\mathcal{D}, \mathcal{V}] \rightarrow [\mathcal{C}, \mathcal{V}]$  is an equivalence then  $\tilde{M}: [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$  is too.*

### 3. THE SETTING

Let  $\mathcal{P}$  be a category with a subcategory  $\mathcal{M}$  containing all the objects.

Assume we have a functor  $(-)^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$  which is the identity on objects and is such that  $m^* \circ m = 1$  for all  $m \in \mathcal{M}$ . In particular, the morphisms in  $\mathcal{M}$  are split monomorphisms (coretractions) in  $\mathcal{P}$ .

We write  $\text{Sub}A$  for the (partially) ordered set of isomorphism classes of morphisms  $m: U \rightarrow A$  in  $\mathcal{M}$ . We will use the term *subobject* rather than “ $\mathcal{M}$ -subobject” for these elements. The order on  $\text{Sub}A$  is the usual one. Abusing notation for this order, we simply write  $U \preceq V$  when there exists an  $f: U \rightarrow V$  such that  $m = n \circ f$ , where  $m: U \rightarrow A$  and  $n: V \rightarrow A$  in  $\mathcal{M}$ . There can be confusion when  $U = V$  as objects, so we assure the reader that we will take care. A subobject of  $A$  is *proper* when it is represented by a non-invertible  $m: U \rightarrow A$  in  $\mathcal{M}$ ; we write  $U \prec A$  or  $U \prec_m A$ .

Define  $\mathcal{R}$  to be the class of morphisms  $r \in \mathcal{P}$  with the property that, if  $r = m \circ x \circ n^*$  with  $m, n \in \mathcal{M}$ , then  $m, n$  are invertible.

Define  $\mathcal{D}$  to be the category with zero morphisms (that is, 1/Set-enriched category) obtained from  $\mathcal{R}$  by adjoining zero morphisms. Composition in  $\mathcal{D}$  of morphisms in  $\mathcal{R}$  is as in  $\mathcal{P}$  if the result is itself in  $\mathcal{R}$ , but zero otherwise.

Define  $\mathcal{S}$  to be the class of morphisms in  $\mathcal{P}$  of the form  $r \circ m^*$  with  $m \in \mathcal{M}$  and  $r \in \mathcal{R}$ .

Recall that the limit of a diagram consisting of a family of morphisms into a fixed object  $A$  is called a *wide pullback*; the morphisms in the limit cone are called *projections*. The dual is *wide pushout*.

**Assumption 3.1.** Wide pullbacks of families of morphisms in  $\mathcal{M}$  exist, have projections in  $\mathcal{M}$ , and become wide pushouts under  $m \mapsto m^*$ .

**Assumption 3.2.** If  $r, r' \in \mathcal{R}$  are composable then  $r' \circ r \in \mathcal{S}$ .

**Assumption 3.3.** If  $r, m^* \circ r \in \mathcal{R}$  and  $m \in \mathcal{M}$  then  $m$  is invertible.

**Assumption 3.4.** The class  $\mathcal{M} \circ \mathcal{M}^*$  of morphisms of the form  $m \circ n^*$  with  $m, n \in \mathcal{M}$  is closed under composition.

**Assumption 3.5.** For all objects  $A \in \mathcal{P}$ , the ordered set  $\text{Sub}A$  is finite.

**Proposition 3.6.** *Any morphism  $f \in \mathcal{P}$  factors as  $f = n \circ r \circ m^*$ , uniquely up to isomorphism for  $m, n \in \mathcal{M}$  and  $r \in \mathcal{R}$ .*

*Proof.* Take  $f: A \rightarrow B$  in  $\mathcal{P}$ . We use Assumption 3.1 twice. Let  $n: Y \rightarrow B$  be the wide pullback of all those morphisms  $V \rightarrow B$  in  $\mathcal{M}$  through which  $f$  factors. Then  $n \in \mathcal{M}$  and there exists a unique  $f_1$  with  $f = n \circ f_1$ . Let  $m: X \rightarrow A$  be the wide pullback of those  $U \rightarrow A$  in  $\mathcal{M}$  whose left adjoint in  $\mathbb{P}$  the morphism  $f_1$  factors through. Then  $m \in \mathcal{M}$  and  $f_1 = r \circ m^*$  for a unique  $r$ . Clearly  $r \in \mathcal{R}$  and we have uniqueness by a familiar argument.  $\square$

**Proposition 3.7.** *If  $t \circ s = m \circ r$  with  $s, t \in \mathcal{S}$ ,  $r \in \mathcal{R}$  and  $m \in \mathcal{M}$  then both  $s$  and  $t$  are in  $\mathcal{R}$ .*

*Proof.* First we prove the weaker form:

*if  $s \circ r = m \circ r'$  with  $r, r' \in \mathcal{R}$ ,  $s \in \mathcal{S}$  and  $m \in \mathcal{M}$  then  $s \in \mathcal{R}$ .*

In obvious notation, put  $s = r_1 \circ m_1^*$ ,  $m_1^* \circ r = m_2 \circ r_2 \circ m_3^*$  and  $r_1 \circ m_2 = m_4 \circ r_3 \circ m_5^*$ . Then  $m \circ r' = s \circ r = r_1 \circ m_1^* \circ r = r_1 \circ m_2 \circ r_2 \circ m_3^* = m_4 \circ r_3 \circ m_5^* \circ r_2 \circ m_3^*$ . Let  $p, q \in \mathcal{M}$  be the projections in the pullback of  $m, m_4$ . Then there exists a unique  $u$  into the pullback with  $r' = p \circ u$  and  $r_3 \circ m_5^* \circ r_2 \circ m_3^* = q \circ u$ . Since  $r'$  can factor through no proper  $\mathcal{M}$ , we deduce that  $p$  is invertible. Put  $n = q \circ p^{-1}$ . Then  $m = m_4 \circ n$  and  $n \circ r' = q \circ u = r_3 \circ m_5^* \circ r_2 \circ m_3^*$ . Therefore  $r' = n^* \circ r_3 \circ m_5^* \circ r_2 \circ m_3^*$ ; by definition of  $\mathcal{R}$ , we obtain that  $m_3$  is invertible. Then  $m_1^* \circ (r \circ m_3) = m_2 \circ r_2$  implies  $(m_1 \circ m_2)^* \circ (r \circ m_3) = r_2$ . Assumption 3.3 applies to yield that  $m_1 \circ m_2$  is invertible. So  $m_1$  has a right inverse, as well as its left inverse  $m_1^*$ , and so is invertible. So  $s = r_1 \circ m_1^*$  is in  $\mathcal{R}$  as asserted.

Now we come to the proof of the Proposition. Since  $s \in \mathcal{S}$ ,  $s = r_1 \circ m_1^*$ . Then, by definition of  $\mathcal{R}$ ,  $m^* \circ t \circ r_1 \circ m_1^* = m^* \circ t \circ s = m^* \circ m \circ r = r$  implies  $m_1$  invertible. So  $s \in \mathcal{R}$ . Now the weaker form above applies to yield  $t \in \mathcal{R}$ .  $\square$

For each  $u: A \rightarrow B$  in  $\mathcal{P}$ , put  $u = m_u \circ s_u$  where  $m_u \in \mathcal{M}$  and  $s_u \in \mathcal{S}$ ; we are using Proposition 3.6.

**Proposition 3.8.** *Suppose  $u: A \rightarrow B$  and  $v: B \rightarrow C$  in  $\mathcal{P}$ .*

- (a) *If  $s_{vu} \in \mathcal{R}$  then  $s_u \in \mathcal{R}$ .*
- (b) *If  $s_{vu} \in \mathcal{R}$  and  $u \in \mathcal{S}$  then  $u, s_v \in \mathcal{R}$ .*

*Proof.* We prove (b) first. We have  $v \circ u = m_v \circ s_v \circ u = m_v \circ m_{s_v u} \circ s_{s_v u}$ . By uniqueness of factorization, we may take  $s_{vu} = s_{s_v u}$ . So  $s_v \circ u = m_{s_v u} \circ s_{vu}$ . By Proposition 3.7, if  $s_{vu} \in \mathcal{R}$  and  $u \in \mathcal{S}$  then  $s_v, u \in \mathcal{R}$ . Now we prove (a). We have  $v \circ u = v \circ m_u \circ s_u$ . By (b),  $s_{vm_u s_u} = s_{vu} \in \mathcal{R}$  implies  $s_u \in \mathcal{R}$ .  $\square$

#### 4. BASIC EXAMPLES

**Example 4.1.** Take a category  $\mathcal{A}$  with a factorization system  $(\mathcal{E}, \mathcal{M})$  in the sense of [14]. Assume that the pullback of any morphism with one in  $\mathcal{M}$  exists. Assume every morphism in  $\mathcal{M}$  is a monomorphism and every object of  $\mathcal{A}$  has only finitely many  $\mathcal{M}$ -subobjects. A span  $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$  is called a *partial map*  $f: X \rightarrow Y$  when  $f_0$  is in  $\mathcal{M}$ . If  $g = (X \xleftarrow{g_0} V \xrightarrow{g_1} Y)$  is another partial map, we write  $f \leq g$  when there exists  $h: U \rightarrow V$  with  $g_0 \circ h = f_0$  and  $g_1 \circ h = f_1$ . Let  $\mathcal{P} = \text{Par}\mathcal{A}$  denote the category whose objects are all those of  $\mathcal{A}$  and whose morphisms are isomorphism classes  $[f]$  of partial maps. Composition is that of spans: that is, by pullback. Taking the usual order  $f \leq g$  on partial maps, we can regard  $\text{Par}\mathcal{A}$  as a locally partially ordered 2-category. We identify  $f: X \rightarrow Y$  in  $\mathcal{A}$  with the morphism  $[1_X, X, f]: X \rightarrow Y$  in  $\mathcal{P}$ . In this way, we have the  $\mathcal{M}$  we require for  $\mathcal{P}$  as the one in  $\mathcal{A}$ . For  $m: U \rightarrow X$  in  $\mathcal{M}$ , a right adjoint in  $\text{Par}\mathcal{A}$  is defined by  $m^* = [m, U, 1_U]: X \rightarrow U$ , and clearly  $m^* \circ m = 1$ . This gives our functor  $(-)^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ . Every partial map  $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$  has

$$[f] = f_1 \circ f_0^*$$

where  $f_0 \in \mathcal{M}$ . Furthermore,  $f_1 = m \circ e$  uniquely up to isomorphism for  $m \in \mathcal{M}$  and  $e \in \mathcal{E}$ . It follows therefore that  $\mathcal{R} = \mathcal{E}$  and that  $[f] \in \mathcal{S}$  if and only if  $f_1 \in \mathcal{E}$ .

Now we look at our Assumptions. To see that Assumption 3.1 holds, first note that since we are assuming the  $\mathcal{M}$ -subobjects form a finite set, finite wide pullbacks can be obtained from pullbacks. By assumption, pullbacks of  $\mathcal{M}$ s exist in  $\mathcal{A}$  and a pullback of an  $\mathcal{M}$  is an  $\mathcal{M}$  in a factorization system. It is a pleasant exercise to see that these pullbacks remain pullbacks in  $\mathcal{P}$  and become pushouts in  $\mathcal{P}$  on taking left adjoints.

We certainly have Assumption 3.2; indeed  $\mathcal{R}$  is closed under composition because  $\mathcal{E}$  is.

For Assumption 3.3, take  $r = [1_A, A, e]$  with  $e \in \mathcal{E}$  and  $m = [1_V, V, m]$  in  $\mathcal{M}$ . To have  $m^* \circ r = [n, P, u] \in \mathcal{R}$ , we must have  $n$  invertible and  $u \in \mathcal{E}$ . Then  $e = m \circ u \circ n^{-1}$  implies  $m \in \mathcal{E}$ ; so  $m$  is invertible.

The class of partial maps in Assumption 3.4 are those of the form  $[m, U, n]$  with  $m, n \in \mathcal{M}$ ; these are closed under composition since pullbacks of  $\mathcal{M}$ s exist and are in  $\mathcal{M}$ .

Assumption 3.5 was one of our assumptions on the factorization system on  $\mathcal{A}$ .

Notice that  $\mathcal{S}$  is not closed under composition unless each pullback of an  $\mathcal{E}$  along an  $\mathcal{M}$  is an  $\mathcal{E}$ . This is true in many examples.

**Example 4.2.** Take  $\mathcal{P}$  to be the category  $\Delta_{\perp, \top}$  of finite non-empty ordinals  $n = \{0, 1, \dots, n-1\}$  with morphisms those functions which preserve first element, last element and order. Functors out of this category are augmented simplicial objects because of the isomorphism (1.2). Morphisms  $\xi, \zeta : m \rightarrow n$  are ordered by taking  $\xi \leq \zeta$  iff and only if  $\xi(i) \leq \zeta(i)$  for all  $i \in m$ . This makes  $\Delta_{\perp, \top}$  into a locally partially ordered 2-category; it is a locally full sub-2-category of  $\text{Cat}$ . Take  $\mathcal{M}$  to consist of all the injective functions in  $\Delta_{\perp, \top}$ ; each such injection  $\partial$  has a left adjoint  $\partial^*$  (and also a right adjoint for that matter) in  $\Delta_{\perp, \top}$ ; clearly  $\partial^*$  is surjective with  $\partial^* \circ \partial = 1$  since  $\partial$  is a fully faithful functor. A surjection  $\sigma$  in  $\mathcal{P}$  is of the form  $\partial^*$  if and only if  $\sigma(i) = 0$  implies  $i = 0$ . We write  $\sigma_k : m+1 \rightarrow m$  for the order-preserving surjection which takes the value  $k$  twice. We write  $\partial_k : m \rightarrow m+1$  for the order-preserving injection which does not have  $k$  in its image. Note that  $\partial_0 \notin \mathcal{P}$ , while  $\sigma_k \dashv \partial_k \dashv \sigma_{k-1}$  in  $\mathbb{P}$  and  $\sigma_k$  is a  $\partial^*$  if and only if  $k > 0$ . Every  $\xi \in \mathcal{P}$  factors uniquely as

$$\xi = \partial_{i_s} \circ \dots \circ \partial_{i_1} \circ \sigma_{j_1} \circ \dots \circ \sigma_{j_t}$$

for  $0 < i_1 < \dots < i_s < n-1$  and  $0 \leq j_1 < \dots < j_t < m-1$ .

We claim  $\mathcal{R}$  consists of the identities and the surjections  $\sigma_0 : m+1 \rightarrow m$ . The only invertible morphisms in  $\mathcal{P}$  are identities. Since members of  $\mathcal{R}$  factor through no proper injection, they must be surjective. Every surjection  $\tau$  is either of the form  $\partial^*$  or uniquely of the form  $\sigma_0 \circ \partial^*$ . Neither of these forms is permissible for  $\tau \in \mathcal{R}$  unless the injection  $\partial$  is an identity. This proves our claim. It is also clear then that  $\mathcal{S}$  consists of all the surjections in  $\mathcal{P}$ .

To prove the Assumptions, we make use of the simplicial identities (see page 24 of [15] for example) which, apart from  $\sigma_i \circ \partial_i = 1 = \sigma_{i-1} \circ \partial_i$ , say that, for all  $i < j$ ,

$$\partial_j \circ \partial_i = \partial_i \circ \partial_{j-1} \text{ , } \sigma_{j-1} \circ \sigma_i = \sigma_i \circ \sigma_j \text{ , } \sigma_j \circ \partial_i = \partial_i \circ \sigma_{j-1} \text{ , } \sigma_i \circ \partial_{j+1} = \partial_j \circ \sigma_i \text{ .}$$

Assumption 3.1 follows from the fact that the pullback of any two monomorphisms in  $\mathcal{P}$  exists and is absolute (that is, preserved by all functors); see page 27 of [15] on

the Eilenberg-Zilber Theorem. Alternatively, notice that the squares

$$\begin{array}{ccc}
 n-1 & \xrightarrow{\partial_{j-1}} & n \\
 \partial_i \downarrow & & \downarrow \partial_i \\
 n & \xrightarrow{\partial_j} & n+1
 \end{array}
 \qquad
 \begin{array}{ccc}
 n-1 & \xleftarrow{\sigma_{j-1}} & n \\
 \sigma_i \uparrow & & \uparrow \sigma_i \\
 n & \xleftarrow{\sigma_j} & n+1
 \end{array}$$

are respectively a pullback and pushout in  $\mathcal{P}$  for  $0 < i < j < n$ . Then the general result follows by stacking these squares vertically and horizontally.

In the present example,  $\mathcal{S}$  is closed under composition, making Assumption 3.2 clear.

For Assumption 3.3, suppose we have  $\partial \circ \rho \in \mathcal{R}$  and  $\rho \in \mathcal{R}$ . Since  $\rho$  is surjective,  $\partial^* \circ \rho = 1$  implies  $\rho = 1$  and hence  $\partial = 1$ , as required. Otherwise  $\partial^* \circ \rho = \sigma_0$ . Yet, if  $\rho = 1$  this contradicts the lack of right adjoint for  $\sigma_0$ . So  $\rho = \sigma_0$ . Thus  $\partial^* \circ \sigma_0 = \sigma_0$ , and we can cancel  $\sigma_0$  to obtain again  $\partial = 1$ .

For Assumption 3.4, the class  $\mathcal{M} \circ \mathcal{M}^*$  of morphisms consists of those which reflect 0. That is,  $\xi = \mu \circ \partial^*$  with  $\mu \in \mathcal{M}$  if and only if  $\xi(i) = 0$  implies  $i = 0$ . This class is clearly closed under composition.

Finally, Assumption 3.5 is clear.

Notice that the arguments above equally apply to the full subcategory  $\Delta_{\perp \neq \top}$  of  $\Delta_{\perp, \top}$  obtained by removing the object 1. Functors  $\Delta_{\perp \neq \top} \rightarrow \mathcal{X}$  are the traditional simplicial objects in  $\mathcal{X}$ .

**Example 4.3.** This example is about the cubical category  $\mathbb{I}$  as used by Sjoerd Crans [11] and Dominic Verity [38, 39]. Functors with domain  $\mathbb{I}$  are cubical objects in the codomain category. Verity constructed  $\mathbb{I}$  as the free monoidal category containing a cointerval.

For each natural number  $k$ , define a poset  $\langle k \rangle = \{-, 1, 2, \dots, k, +\}$  by adjoining a bottom element  $-$  and a top element  $+$  to the discrete poset  $\{1, 2, \dots, k\}$ . Any function  $f : \langle k \rangle \rightarrow \langle h \rangle$  which preserves top and bottom is order-preserving. Thus we get a locally partially ordered 2-category with objects the  $\langle k \rangle$ , with morphisms the top-and-bottom-preserving functions, and with the pointwise order. Let  $\mathbb{I}$  be the locally full sub-2-category consisting of those  $f : \langle k \rangle \rightarrow \langle h \rangle$  for which, if  $f(i), f(j) \notin \{-, +\}$  then  $i < j$  if and only if  $f(i) < f(j)$ .

Let  $\mathcal{P}$  be the underlying category of this  $\mathbb{I}$ . Let  $\mathcal{M}$  consist of the morphisms in  $\mathcal{P}$  which are injective as functions. Given such an  $m : \langle k \rangle \rightarrow \langle h \rangle$  in  $\mathcal{M}$ , define  $m^* : \langle h \rangle \rightarrow \langle k \rangle$  to send each  $m(i)$  in the image of  $m$  to  $i$  and everything else to  $+$ . Clearly  $m^* \in \mathcal{P}$  and  $m^* \circ m = 1$ . Furthermore,  $mm^*(j)$  is equal to  $j$  if  $j = m(i)$  for some  $i$ , and  $+$  otherwise. Therefore  $1 \leq m \circ m^*$  showing  $m^*$  to be left adjoint to  $m$  with identity counit.

We can characterize morphisms of the form  $m^*$  as those which are surjective as functions and reflect the bottom element  $-$ . Consequently  $\mathcal{R}$  consists of the morphisms which are surjective as functions and reflect the top element  $+$ .

Assumption 3.5 and 3.2 are clear.

$$\begin{array}{ccc}
 \langle \ell \rangle & \xrightarrow{q} & \langle v \rangle \\
 p \downarrow & & \downarrow n \\
 \langle u \rangle & \xrightarrow{m} & \langle k \rangle
 \end{array}
 \quad
 \begin{array}{ccc}
 \langle k \rangle & \xrightarrow{n^*} & \langle v \rangle \\
 m^* \downarrow & & \downarrow q^* \\
 \langle u \rangle & \xrightarrow{p^*} & \langle \ell \rangle
 \end{array}
 \quad
 \begin{array}{c}
 \langle v \rangle \searrow g \\
 \langle u \rangle \xrightarrow{p^*} \langle \ell \rangle \xrightarrow{f} \langle h \rangle
 \end{array}
 \quad (4.8)$$

For Assumption 3.1, the existence of intersections is obvious. However, we must show that taking left adjoints gives cointersections. Take a pullback as in the left-hand diagram of (4.8) and consider the right-hand diagram. Assume  $f \circ m^* = g \circ n^*$ . It suffices to show  $f \circ p \circ p^* = f$ . Now  $fpp^*(i)$  is equal to  $fp(j)$  if  $i = p(j)$  for some  $j$ , and equal to  $+$  otherwise. In the first case, we have  $fpp^*(i) = fp(j) = f(i)$ , as required. In the second case, if  $i$  does not have the form  $p(j)$  then  $m(i)$  is not in the image of  $n$ , and so  $gn^*m(i) = +$ ; thus  $f(i) = fm^*m(i) = +$  and we again have  $fpp^*(i) = f(i)$ .

For Assumption 3.3, suppose  $m \in \mathcal{M}$  and  $r, m^* \circ r \in \mathcal{R}$ . Suppose  $m^*(i) = +$ . Since  $r$  is surjective, we have  $i = r(j)$  so that  $m^*r(j) = +$ . However  $m^* \circ r$  reflects  $+$ . So  $j = +$ , yielding  $i = r(+) = +$ . This proves  $m^*$  reflects  $+$  and therefore must be invertible.

For Assumption 3.4, the composites of the form  $mon^*$  are clearly the (not necessarily surjective) morphisms which reflect  $-$ . Clearly these are closed under composition.

There is another possible characterization of  $\mathcal{R}$ , namely as the category of right adjoints to the morphisms in  $\mathcal{M}$ . Thus in fact  $\mathcal{R}$  is dual to  $\mathcal{M}$ . Now  $\mathcal{M}$  is really just the category  $\Delta_{\text{inj}}$  of finite ordinals and injective order-preserving maps, and  $\mathcal{R} \cong \mathcal{M}^{\text{op}}$ . Also the category  $\mathcal{M}^*$ , with morphisms the  $m^* \in \mathcal{M}$ , is dual to  $\mathcal{M}$ . The factorization of Proposition 3.6 in this case shows the category  $\mathcal{P}$  is a composite

$$\mathbb{I} = \Delta_{\text{inj}} \circ \Delta_{\text{inj}}^{\text{op}} \circ \Delta_{\text{inj}}^{\text{op}},$$

relative to suitably defined distributive laws.

An alternative viewpoint is that  $\mathbb{I}$  is  $\text{ParPar}\Delta_{\text{inj}}$ , where in each case partial maps are defined relative to the morphisms in  $\Delta_{\text{inj}}$ .

## 5. THE KERNEL MODULE

In the setting of Section 3, we shall define a functor

$$M: \mathcal{D}^{\text{op}} \times \mathcal{P} \longrightarrow \text{Agp} \quad (5.9)$$

which preserves zeros in the first variable.

Let  $\mathbb{Z}\Lambda$  denote the free abelian group on the set  $\Lambda$ . This defines the value on objects of a strong monoidal functor  $\mathbb{Z}: \text{Set} \rightarrow \text{Agp}$  whose right adjoint is the forgetful functor.

For each  $u: A \rightarrow B$  in  $\mathcal{P}$ , we have  $m_u \in \mathcal{M}$  and  $s_u \in \mathcal{S}$  as in the triangle

$$\begin{array}{ccc}
 A & \xrightarrow{u} & B \\
 s_u \searrow & & \nearrow m_u \\
 & S_u &
 \end{array}
 \quad (5.10)$$

by using Proposition 3.6.

Define  $M$  on objects by

$$M(A, B) = \mathbb{Z}\{u \in \mathcal{P}(A, B) : s_u \in \mathcal{R}\} . \quad (5.11)$$

Suppose  $r : A_1 \rightarrow A$  in  $\mathcal{R}$  and  $f : B \rightarrow B_1$  in  $\mathcal{P}$ . Define  $M$  on morphisms by

$$M(r, f)u = \begin{cases} f \circ u \circ r & \text{for } s_{fur} \in \mathcal{R} \\ 0 & \text{otherwise} . \end{cases} \quad (5.12)$$

To verify that this is a functor we need to see that

$$M(r_1, f_1)M(r, f)u = M(r \circ_{\mathcal{D}} r_1, f_1 \circ f)u$$

holds. This amounts to showing that the left-hand side is non-zero if and only if the right-hand side is, because then both sides equal  $f_1 \circ f \circ u \circ r \circ r_1$ . In other words, we need to see that  $s_{fur}, s_{f_1 f_{urr_1}} \in \mathcal{R}$  if and only if  $r \circ r_1, s_{f_1 f_{urr_1}} \in \mathcal{R}$ . In fact,  $s_{f_1 f_{urr_1}} \in \mathcal{R}$  implies both  $s_{fur} \in \mathcal{R}$  and  $r \circ r_1 \in \mathcal{R}$ . For we know by Assumption 3.2 that  $r \circ r_1 \in \mathcal{S}$ ; so, by Proposition 3.8(b),  $s_{f_1 f_{urr_1}} \in \mathcal{R}$  implies  $r \circ r_1 \in \mathcal{R}$ . By Proposition 3.8(a), we also conclude that  $s_{f_{urr_1}} \in \mathcal{R}$ ; then Proposition 3.8(b) gives  $s_{fur} \in \mathcal{R}$ .

Let  $\mathbb{Z}_\dagger \mathcal{D}$  denote the free Agp-enriched category on the pointed-set-enriched category  $\mathcal{D}$ . Let  $\mathbb{Z}_* \mathcal{P}$  denote the free Agp-enriched category on the category  $\mathcal{P}$ . For any additive (that is, Agp-enriched) category  $\mathcal{X}$ , we have isomorphisms of categories

$$[\mathbb{Z}_\dagger \mathcal{D}, \mathcal{X}]_{\text{add}} \cong [\mathcal{D}, \mathcal{X}]_{\text{pt}} \quad \text{and} \quad [\mathbb{Z}_* \mathcal{P}, \mathcal{X}]_{\text{add}} \cong [\mathcal{P}, \mathcal{X}] ,$$

where, for emphasis, we write the subscript “pt” for the pointed-set-enriched functor category and “add” for the Agp-enriched functor category.

We identify (5.9) with its obvious extension to an additive functor

$$M : \mathbb{Z}_\dagger \mathcal{D}^{\text{op}} \otimes \mathbb{Z}_* \mathcal{P} \longrightarrow \text{Agp} . \quad (5.13)$$

Now we are in the situation for a kernel adjunction of the form (2.5) with  $\mathcal{V} = \text{Agp}$ . We have an adjunction.

$$[\mathcal{D}, \text{Agp}]_{\text{pt}} \begin{array}{c} \xrightarrow{\widehat{M}} \\ \perp \\ \xleftarrow{\widetilde{M}} \end{array} [\mathcal{P}, \text{Agp}] . \quad (5.14)$$

More combinatorial expressions than (2.7) and (2.6) exist for  $\widehat{M}$  and  $\widetilde{M}$  in our present situation.

**Theorem 5.1.** *Suppose  $M$  is the kernel as in (5.13). For any zero preserving functor  $F : \mathcal{D} \rightarrow \text{Agp}$  and any functor  $T : \mathcal{P} \rightarrow \text{Agp}$ , there are bijections*

$$\widehat{M}(F)B \cong \sum_{S \preceq_m B} FS , \quad \widetilde{M}(T)A \cong \bigcap_{U \prec_m A} \ker T(m^* : A \rightarrow U) .$$

*Proof.* We shall prove

$$\int^{A \in \mathbb{Z}_\dagger \mathcal{D}} M(A, B) \otimes FA \cong \sum_{S \preceq_m B} FS$$

by showing that the family of morphisms

$$\chi_A : M(A, B) \otimes FA \longrightarrow \sum_{S \preceq_m B} FS$$

defined by  $\chi_A(u \otimes x) = (Fs_u)x \in FS_u$  (using the notation of (5.10)) is a universal extraordinary natural transformation (that is, a universal wedge). In order for a family  $\theta_A: M(A, B) \otimes FA \rightarrow X$  in  $\mathbf{Agp}$  to be extraordinary natural, we require

$$\theta_A(u \otimes (Fr)y) = \begin{cases} \theta_{A_1}((u \circ r) \otimes y) & \text{for } s_{ur} \in \mathcal{R} \\ 0 & \text{otherwise} . \end{cases} \quad (5.15)$$

for all  $r: A_1 \rightarrow A$  in  $\mathcal{R}$ ,  $u: A \rightarrow B$  in  $\mathcal{P}$  with  $s_u \in \mathcal{R}$ , and  $y \in FA_1$ . Notice that in this situation  $s_{ur} = s_u \circ r$  by uniqueness of factorization and Assumption 3.2. Now for the family  $\chi$  we have  $\chi_A(u \otimes (Fr)y) = F(s_u \circ_{\mathcal{D}} r)y$ . If  $s_{ur} \in \mathcal{R}$  then  $s_u \circ_{\mathcal{D}} r = s_u \circ r = s_{ur}$ , so  $F(s_u \circ_{\mathcal{D}} r)y = F(s_{ur})y = \chi_{A_1}((u \circ r) \otimes y)$ . Otherwise,  $F(s_u \circ_{\mathcal{D}} r) = F(0) = 0$ , as required. For universality, we must see that a general wedge  $\theta$  factors uniquely through  $\chi$ . Define  $\phi: \sum_{S \preceq_m B} FS \rightarrow X$  by  $\phi(x \in FS) = \theta_S(m \otimes x)$ . With  $s_u \in \mathcal{R}$ , we have, using (5.15),

$$\theta_A(u \otimes x) = \theta_A((m_u \circ s_u) \otimes x) = \theta_S(m_u \otimes (Fs_u)x) = \phi((Fs_u)x \in FS_u) = \phi\chi_A(u \otimes x) .$$

Thus  $\theta = \phi\chi$ . By taking  $u = m \in \mathcal{M}$ , we also see that the definition of  $\phi$  is forced.

Define  $\tilde{T}A = \bigcap_{U \prec_m A} \ker T(m^*: A \rightarrow U)$ , and, for  $r: A \rightarrow A_1$ , define  $(\tilde{T}r)a = (Tr)a$ . For this we need to see that  $a \in \tilde{T}A$  implies  $(Tr)a \in \tilde{T}A_1$ . Take a non-invertible  $n: V \rightarrow A_1$ . Using Proposition 3.6, we have  $n^* \circ r = \ell \circ r_1 \circ m^*$ . So  $(n \circ \ell)^* \circ r = r_1 \circ m^*$ . If  $m$  is invertible then  $(n \circ \ell)^* \circ r \in \mathcal{R}$ . By Assumption 3.3,  $n \circ \ell$  is invertible. So  $n$  is invertible, a contradiction. So  $m$  is not invertible and we have  $(Tn^*)(Tr)a = T(\ell)T(r_1)(Tm^*)a = T(\ell)T(r_1)0 = 0$ . So  $(Tr)a \in \tilde{T}A_1$ .

Put  $\hat{F}B = \sum_{S \preceq_m B} FS$  and transport the functoriality in  $B$  across the isomorphism with  $\widehat{M}(F)B$ . We will prove that there is a natural isomorphism

$$[\mathcal{P}, \mathcal{X}](\hat{F}, T) \cong [\mathcal{D}, \mathcal{X}]_{\text{pt}}(F, \tilde{T})$$

and hence conclude that  $\widehat{M}(T) \cong \tilde{T}$ . Take a natural transformation  $\theta: \hat{F} \Rightarrow T: \mathcal{P} \rightarrow \mathcal{X}$ . The definition of  $\hat{F}f$  involves writing  $f \circ m = m_{fm} \circ s_{fm}$  then, with  $s_{fm} \in \mathcal{R}$ , we have commutativity in the following diagram.

$$\begin{array}{ccccc} FU & \xrightarrow{\text{in}_U} & \hat{F}A & \xrightarrow{\theta_A} & TA \\ \downarrow F s_{fm} & & \downarrow \hat{F}f & & \downarrow Tf \\ FV & \xrightarrow{\text{in}_V} & \hat{F}B & \xrightarrow{\theta_B} & TB \end{array} \quad (5.16)$$

Consider a noninvertible  $\ell: W \rightarrow A$  in  $\mathcal{M}$ . Then, by the properties of  $\mathcal{R}$ ,  $\ell^* \in \mathcal{S}$  and  $\ell^* \notin \mathcal{R}$ . So  $T\ell^* \circ \theta_A \circ \text{in}_A = \theta_W \circ \hat{F}\ell^* \circ \text{in}_A = \theta_W \circ 0 = 0$ . This implies there exists a unique morphism  $\phi_A: FA \rightarrow \tilde{T}A$  such that the following square commutes.

$$\begin{array}{ccc} FA & \xrightarrow{\phi_A} & \tilde{T}A \\ \text{in}_A \downarrow & & \downarrow i_A \\ \hat{F}A & \xrightarrow{\theta_A} & TA \end{array} \quad (5.17)$$

Naturality of  $\phi$  is proved as follows using (5.16) with  $f = r \in \mathcal{R}$ :  $i_B \circ \phi_B \circ Fr = \theta_B \circ \text{in}_B \circ Fr = \theta_B \circ \hat{F}r \circ \text{in}_A = Tr \circ \theta_A \circ \text{in}_A = Tr \circ i_A \circ \phi_A = i_B \circ \tilde{T}r \circ \phi_A$ .

For the inverse direction, take any natural transformation  $\phi: F \Rightarrow \tilde{T}: \mathcal{D} \rightarrow \mathcal{X}$ . Define  $\theta$  by commutativity of the following diagram.

$$\begin{array}{ccccc} FU & \xrightarrow{\text{in}_U} & \widehat{F}A & \xrightarrow{\theta_A} & TA \\ \phi_U \downarrow & & & & \uparrow Tm \\ \tilde{T}U & \xrightarrow{i_U} & TU & & \end{array} \quad (5.18)$$

We need to prove the right-hand square of (5.16) commutes when precomposed with any  $\text{in}_U$  for any  $m: U \rightarrow A$  in  $\mathcal{M}$ . In the case where  $s_{fm} \in \mathcal{R}$ , the desired commutativity is a consequence of the commutativity of the following three squares.

$$\begin{array}{ccccccc} FU & \xrightarrow{\phi_U} & \tilde{T}U & \xrightarrow{i_A} & TU & \xrightarrow{Tm} & TA \\ F s_{fm} \downarrow & & \tilde{T} s_{fm} \downarrow & & \downarrow T s_{fm} & & \downarrow T f \\ FV & \xrightarrow{\phi_V} & \tilde{T}V & \xrightarrow{i_B} & TV & \xrightarrow{T n_{fm}} & TB \end{array} \quad (5.19)$$

In the case where  $s_{fm} \notin \mathcal{R}$ , we can write  $s_{fm} = r \circ \ell^*$  for some noninvertible  $\ell \in \mathcal{M}$ . Then (using both  $U \preceq A$  and  $U \preceq U$ ) we have  $Tf \circ \theta_A \circ \text{in}_U = T(f \circ m) \circ i_U \circ \phi_U = T(n_{fm} \circ r) \circ T\ell^* \circ i_U \circ \phi_U = T(n_{fm} \circ r) \circ 0 \circ \phi_U = 0 = \theta_B \circ \widehat{F}f \circ \text{in}_U$ , as required.

To show that the assignments are mutually inverse, take  $\theta$  and define  $\phi$  by (5.17). Let  $\bar{\theta}$  be as  $\theta$  is in (5.18). Then  $\bar{\theta}_A \circ \text{in}_U = Tm \circ i_U \circ \phi_U = Tm \circ \theta_U \circ \text{in}_U = \theta_A \circ \widehat{F}m \circ \text{in}_U = \theta_A \circ \text{in}_U$ . So  $\bar{\theta} = \theta$ .

On the other hand, take  $\phi$  and define  $\theta$  by (5.18). Let  $\phi'$  be as  $\phi$  is in (5.17). Then  $i_A \circ \phi'_A = \theta_A \circ \text{in}_A = i_A \circ \phi_A$ . So  $\phi' = \phi$ .  $\square$

## 6. REDUCTION OF THE PROBLEM

The problem in question is to show the adjunction (5.14) is an equivalence. We wish to use the theory in Section 2 to reduce the problem to a special case of Example 4.1 involving partial maps. In Section 7, we will treat the general case of Example 4.1.

Consider  $\mathcal{P}, \mathcal{M}, \mathcal{I}, \mathcal{R}$  and  $\mathcal{D}$  as in the setting of Section 3. Let  $\mathcal{I}$  be the class of invertible morphisms in  $\mathcal{P}$ . Let  $\mathcal{K} = \mathcal{M} \circ \mathcal{M}^*$  be the subcategory of  $\mathcal{P}$  as assured by Assumption 3.4. Then  $(\mathcal{I}, \mathcal{M})$  is a factorization system on  $\mathcal{M}$  becoming an instance of Example 4.1 with  $\text{Par}\mathcal{M} = \mathcal{K}$ . Denote the kernel module for this situation by  $N: \mathcal{I}^{\text{op}} \times \mathcal{K} \rightarrow \text{Agp}$ ; on objects it is given by  $N(C, D) = \mathbb{Z}\mathcal{M}(C, D)$ .

Let  $H: \mathcal{I} \rightarrow \mathcal{D}$  and  $K: \mathcal{K} \rightarrow \mathcal{P}$  be the inclusions; they are the identity on objects and so Cauchy dense.

We have the obvious inclusion

$$\theta_{C,D}: N(C, D) \rightarrow M(HC, KD) \quad (6.20)$$

taking  $m \in \mathcal{M}(C, D)$  to  $m \in \mathcal{P}(C, D)$  (using  $s_m = 1$ ).

**Proposition 6.1.** *For  $\theta: N \Rightarrow M(H, K)$  defined at (6.20), the corresponding natural families*

$$\theta_{A,D}^r: \int^{C \in \mathcal{I}} N(C, D) \otimes \mathcal{D}(A, HC) \longrightarrow M(A, KD) ,$$

$$\theta_{C,B}^\ell: \int^{D \in \mathcal{K}} \mathcal{P}(KD, B) \cdot N(C, D) \longrightarrow M(HC, B),$$

as in Section 2, are both invertible.

*Proof.* For  $r: A \rightarrow C$  in  $\mathcal{R}$  and  $m: C \rightarrow D$  in  $\mathcal{M}$ , we have  $\theta_{A,D}^r[m \otimes r] = m \circ r$ . Yet every  $u: A \rightarrow D$  with  $s_u \in \mathcal{R}$  factors uniquely up to a morphism in  $\mathcal{I}$  as  $u = m_u \circ s_u = \theta_{A,D}^r[m_u \otimes s_u]$ . So we have our inverse bijection.

By the definition of the second coend, for all  $k: D \rightarrow D_1$  in  $\mathcal{K}$  and  $g: D_1 \rightarrow B$ , we have  $[(g \circ k) \otimes m] = [(g \otimes k) \circ m]$  when  $k \circ m \in \mathcal{M}$ , zero otherwise. For  $m: C \rightarrow D$  in  $\mathcal{M}$  and  $f: D \rightarrow B$  in  $\mathcal{P}$ , we have  $[f \otimes m] = [(f \circ m) \otimes 1] = [u \otimes 1]$  for some  $u \in \mathcal{P}$ . But  $u = \ell \circ r \circ n^*$  for some  $n, \ell \in \mathcal{M}$  and  $r \in \mathcal{R}$ . So  $[u \otimes 1] = [(\ell \circ r) \otimes n^*]$  provided  $n^* \in \mathcal{M}$ , zero otherwise. However,  $n^* \in \mathcal{M}$  implies  $n$  invertible. So  $s_u \in \mathcal{R}$ . In other words,  $[f \otimes m]$  is either zero or of the form  $[u \otimes 1]$  for a unique  $u$  with  $s_u \in \mathcal{R}$ . By definition,  $\theta_{C,B}^\ell[f \otimes m] = f \circ m$  when  $s_{fm} \in \mathcal{R}$ , zero otherwise. We have shown that an inverse to  $\theta_{C,B}^\ell$  is provided by  $u \mapsto [u \otimes 1_C]$ .  $\square$

Combining Proposition 6.1 and Corollary 2.2, we have:

**Corollary 6.2.** *If  $\tilde{N}$  is an equivalence then so is  $\tilde{M}$ .*

## 7. A TREATMENT OF THE PARTIAL MAP CASE

We begin with any category  $\mathcal{A}$  equipped with a factorization system  $(\mathcal{E}, \mathcal{M})$  as in Example 4.1. We have  $\mathcal{P} = \text{Par}\mathcal{A}$  and  $\mathcal{R} = \mathcal{E}$ . Also  $\mathcal{D}$  is  $\mathcal{E}$  with adjoined zero morphisms, so  $[\mathcal{D}, \mathcal{X}]_{\text{pt}} \cong [\mathcal{E}, \mathcal{X}]$  for any category  $\mathcal{X}$  with zero morphisms.

For every pullback

$$\begin{array}{ccc} P & \xrightarrow{g} & B \\ n \downarrow & & \downarrow m \\ A & \xrightarrow{f} & C \end{array} \quad (7.21)$$

in  $\mathcal{A}$  with  $m \in \mathcal{M}$ , the square

$$\begin{array}{ccc} P & \xrightarrow{g_*} & B \\ n^* \uparrow & & \uparrow m^* \\ A & \xrightarrow{f_*} & C \end{array} \quad (7.22)$$

commutes in  $\mathcal{P}$ .

There is analogue, for partial maps in place of general spans, of the result in [26] about Mackey functors. There is an equivalence between functors  $T: \mathcal{P} \rightarrow \mathcal{X}$  and pairs  $(T^*, T_*)$  of functors  $T^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{X}$  and  $T_*: \mathcal{A} \rightarrow \mathcal{X}$  which agree on objects and satisfy the Bénabou-Roubaud-Chevalley-Beck condition on pullbacks of the form (7.21). Indeed  $T^* = T \circ (-)^*$  and  $T_* = T \circ (-)_*$ .

Let  $J: \mathcal{E} \rightarrow \mathcal{A}$  be the inclusion functor and let  $I = (-)_*: \mathcal{A} \rightarrow \mathcal{P}$ . Both functors are the identity on objects.

Let  $\mathcal{X}$  be any category admitting coproducts and products indexed by sets of  $\mathcal{M}$ -subobjects of any given object of  $\mathcal{A}$ .

**Lemma 7.1.** *Each functor  $F: \mathcal{E} \rightarrow \mathcal{X}$  has a pointwise left Kan extension*

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{J} & \mathcal{A} \\ & \searrow F & \swarrow \text{Lan}_J F \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xrightarrow{\kappa} \\ \end{array} \quad (7.23)$$

along  $J: \mathcal{E} \rightarrow \mathcal{A}$  defined on objects by:

$$(\text{Lan}_J F)X = \sum_{U \leq X} FU .$$

For morphisms  $f: X \rightarrow Y$  in  $\mathcal{A}$ , the following square commutes.

$$\begin{array}{ccc} FU & \xrightarrow{\text{in}_U} & (\text{Lan}_J F)X \\ \downarrow Fe & & \downarrow (\text{Lan}_J F)f \\ FfU & \xrightarrow{\text{in}_{fU}} & (\text{Lan}_J F)Y \end{array} \quad (7.24)$$

*Proof.* The inclusion  $U \leq X \mapsto (U, i_U: JU \rightarrow X)$  of the discrete category on the set  $\{U: U \leq X\}$  into the comma category  $J/X$  has a left adjoint, taking  $(V, f: JV \rightarrow X)$  to  $fV \leq X$ , and so is final. It follows that the colimit of  $J/X \xrightarrow{\text{dom}} \mathcal{E} \xrightarrow{F} \mathcal{X}$  can be calculated by restricting along the inclusion and so is the coproduct displayed.  $\square$

**Lemma 7.2.** *Each functor  $T: \mathcal{A} \rightarrow \mathcal{X}$  has a pointwise right Kan extension*

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{I} & \mathcal{P} \\ & \searrow T & \swarrow \text{Ran}_I T \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xleftarrow{\rho} \\ \end{array} \quad (7.25)$$

along  $I: \mathcal{A} \rightarrow \mathcal{P}$  defined on objects by:

$$(\text{Ran}_I T)X = \prod_{U \leq X} TU .$$

For morphisms  $(W, h): X \rightarrow Y$  in  $\mathcal{P}$ , the square

$$\begin{array}{ccc} (\text{Ran}_I T)X & \xrightarrow{\text{pr}_{h^{-1}V}} & Th^{-1}V \\ (\text{Ran}_I T)(W, h) \downarrow & & \downarrow T\bar{h} \\ (\text{Ran}_I T)Y & \xrightarrow{\text{pr}_V} & TV \end{array} \quad (7.26)$$

commutes, where  $\bar{h}$  is the pullback of  $h$  along  $i_V$ .

*Proof.* The functor  $U \leq X \mapsto (i_U^*: X \rightarrow IU, U)$  from the discrete category on the set  $\{U: U \leq X\}$  into the comma category  $X/I$  has a right adjoint, taking  $((W, h): X \rightarrow IA, A)$  to  $W \leq X$ , and so is initial. It follows that the limit of  $X/I \xrightarrow{\text{cod}} \mathcal{A} \xrightarrow{T} \mathcal{X}$  can be calculated as the product displayed.  $\square$

This gives the two adjunctions

$$[\mathcal{E}, \mathcal{X}] \xrightleftharpoons[\begin{smallmatrix} \perp \\ [J,1] \end{smallmatrix}]{\text{Lan}_J} [\mathcal{A}, \mathcal{X}] \xrightleftharpoons[\begin{smallmatrix} \top \\ [I,1] \end{smallmatrix}]{\text{Ran}_I} [\mathcal{P}, \mathcal{X}] . \quad (7.27)$$

Each adjunction generates a comonad on  $[\mathcal{A}, \mathcal{X}]$ :

$$G = \text{Lan}_J \circ [J, 1] \quad \text{and} \quad H = [I, 1] \circ \text{Ran}_I . \quad (7.28)$$

To be more explicit, the functors  $GF$  and  $HF$  are defined on objects by

$$(GF)A = \sum_{U \leq A} FU \quad \text{and} \quad (HF)A = \prod_{U \leq A} FU .$$

On morphisms  $f: A \rightarrow B$  in  $\mathcal{A}$ , they are defined by commutativity in the squares

$$\begin{array}{ccc} FU & \xrightarrow{Fe} & FfU \\ \text{in}_U \downarrow & & \downarrow \text{in}_{fU} \\ (GF)A & \xrightarrow{(GF)f} & (GF)B \end{array} \quad \begin{array}{ccc} (HF)A & \xrightarrow{(HF)f} & (HF)B \\ \text{pr}_{f^{-1}V} \downarrow & & \downarrow \text{pr}_V \\ Ff^{-1}V & \xrightarrow{F\bar{f}} & FV \end{array} . \quad (7.29)$$

The counits  $\varepsilon_F: GF \rightarrow F$  and  $\varepsilon_F: HF \rightarrow F$  have components respectively defined by  $\varepsilon_F A \circ \text{in}_U = F\text{in}_U$  and  $\varepsilon_F A = \text{pr}_A$ . The comultiplications  $\delta_F: GF \rightarrow G^2F$  and  $\delta_F: HF \rightarrow H^2F$  have components respectively defined as follows.

$$\begin{array}{ccc} FU & \xrightarrow{\text{in}_U} & \sum_{U \leq A} FU \\ & \searrow \text{in}_{U \leq U} & \swarrow \delta_F A \\ & \sum_{V \leq U \leq A} FV & \end{array} \quad \begin{array}{ccc} \prod_{U \leq A} FU & \xrightarrow{\text{pr}_V} & FV \\ & \searrow \delta_F A & \swarrow \text{pr}_{V \leq U} \\ & \prod_{V \leq U \leq A} FV & \end{array}$$

We shall construct a comonad morphism  $\Theta: G \Rightarrow H$  when  $\mathcal{X}$  is a pointed category. Take  $F: \mathcal{A} \rightarrow \mathcal{X}$  and  $A \in \mathcal{A}$ . We define

$$\Theta_F A: \sum_{U \leq A} FU \rightarrow \prod_{V \leq A} FV \quad (7.30)$$

by taking the composite

$$FU \xrightarrow{\text{in}_U} \sum_{U \leq A} FU \xrightarrow{\Theta_F A} \prod_{V \leq A} FV \xrightarrow{\text{pr}_V} FV$$

to be zero unless  $U \leq V$ , in which case the composite is  $F(U \leq V): FU \rightarrow FV$ .

**Proposition 7.3.** *For the two comonads  $G$  and  $H$  of (7.28) on  $[\mathcal{A}, \mathcal{X}]$ , a comonad morphism  $\Theta: G \Rightarrow H$  is defined by the components (7.30).*

*Proof.* Naturality in  $A$  is proved by contemplating the following diagram.

$$\begin{array}{ccccccc} FU & \xrightarrow{\text{in}_U} & \sum_{U \leq A} FU & \xrightarrow{\Theta_F A} & \prod_{U_1 \leq A} FU_1 & \xrightarrow{\text{pr}_{f^{-1}V}} & Ff^{-1}V \\ \text{Fe} \downarrow & & \downarrow (GF)f & & \downarrow (HF)f & & \downarrow F\bar{f} \\ FfU & \xrightarrow{\text{in}_{fU}} & \sum_{V_1 \leq B} FV_1 & \xrightarrow{\Theta_F B} & \prod_{V \leq B} FV & \xrightarrow{\text{pr}_V} & FV \end{array}$$

The top composite is zero unless  $U \leq f^{-1}V$  (say with inclusion  $i$ ). The bottom composite is zero unless  $fU \leq V$  (say with inclusion  $j$ ). The conditions are the same. When this condition holds, the diagram commutes since  $\bar{f} \circ i = j \circ e$ .

Naturality in  $F$  is obvious.

Preservation of the counits is proved by the calculation

$$\varepsilon_F A \circ \Theta_F A \circ \text{in}_U = \text{pr}_A \circ \Theta_F A \circ \text{in}_U = F \text{in}_U = \varepsilon_F A \circ \text{in}_U .$$

Preservation of the comultiplications is proved by observing that

$$\text{pr}_{V \leq U} \circ \delta_F A \circ \Theta_F A \circ \text{in}_W = \text{pr}_V \circ \Theta_F A \circ \text{in}_W$$

while

$$\text{pr}_{V \leq U} \circ \Theta_F^2 A \delta_F A \circ \text{in}_W = \text{pr}_{V \leq U} \circ \Theta_F^2 A \circ \text{in}_{W \leq W} .$$

Both of these right-hand sides are zero unless  $W \leq V$ , in which case they are both equal to  $F(W \leq V)$ .  $\square$

**Proposition 7.4.** *The functor  $[I, 1] : [\mathcal{P}, \mathcal{X}] \rightarrow [\mathcal{A}, \mathcal{X}]$  is comonadic.*

*Proof.* The functor  $[I, 1]$  is conservative since  $I$  is bijective on objects. It also preserves all limits, including all equalizers. The result follows (for example) by the Beck comonadicity theorem [27].  $\square$

It follows that  $\Theta : G \Rightarrow H$  induces a functor  $\bar{\Theta} : [\mathcal{A}, \mathcal{X}]^G \rightarrow [\mathcal{P}, \mathcal{X}]$  over  $[\mathcal{A}, \mathcal{X}]$ , where  $[\mathcal{A}, \mathcal{X}]^G$  is the category of Eilenberg-Moore  $G$ -coalgebras. The composite of  $\bar{\Theta}$  with the comparison functor  $[\mathcal{E}, \mathcal{X}] \rightarrow [\mathcal{A}, \mathcal{X}]^G$  is isomorphic to

$$\widehat{M}(F) : [\mathcal{E}, \mathcal{X}] \rightarrow [\mathcal{P}, \mathcal{X}] \quad (7.31)$$

as obtained in Theorem 5.1.

Using the finite well-poweredness assumption on the factorization system of  $\mathcal{A}$ , we shall show that the functor (7.31) is an equivalence. In other words, we shall show that the free additive categories on  $\mathcal{E}$  and  $\mathcal{P}$  are Morita equivalent (= additively Cauchy equivalent).

**Theorem 7.5.** *In the situation described in Example 4.1 and for any additive category  $\mathcal{X}$  with finite direct sums and splitting idempotents, the functor (7.31) is an equivalence of categories*

$$[\mathcal{E}, \mathcal{X}] \simeq [\mathcal{P}, \mathcal{X}] .$$

We first prove that the comonads  $G$  and  $H$  of (7.28) are isomorphic.

**Proposition 7.6.** *In the situation of Example 4.1 and  $\mathcal{X}$  as in Theorem 7.5, the comonad morphism  $\Theta : G \Rightarrow H$  defined at (7.30) is invertible.*

*Proof.* The ordered set of  $\mathcal{M}$ -subobjects of  $A \in \mathcal{A}$  is assumed finite and so has a linear refinement. So we can list all the subobjects as  $0 = U_0, U_1, \dots, U_n = A$  such that  $U_i \leq U_j$  implies  $i \leq j$ . Then the morphisms (7.30) are represented by an upper-triangular matrix with identity morphisms in the main diagonal. Since  $\mathcal{X}$  is additive (including existence of additive inverses), such matrices are invertible.  $\square$

*Proof of Theorem 7.5.* By Proposition 7.6, the functor (7.31) becomes the comparison from  $[\mathcal{E}, \mathcal{X}]$  into the category of  $G$ -coalgebras. So the Theorem is now equivalent to comonadicity of the functor  $\text{Lan}_J : [\mathcal{E}, \mathcal{X}] \rightarrow [\mathcal{A}, \mathcal{X}]$ . Since the unit of the adjunction

$\text{Lan}_J \dashv [J, 1]$  is a pointwise coretraction  $FA \longrightarrow FA \oplus \bigoplus_{U < A} FU$ , and hence a strong monomorphism, the functor  $\text{Lan}_J$  is conservative. So it remains to prove that  $\text{Lan}_J$  preserves certain equalizers. In fact, it preserves all finite limits. Since limits in  $[\mathcal{A}, \mathcal{X}]$  are formed pointwise, it suffices to see that each

$$\text{ev}_A \circ \text{Lan}_J : [\mathcal{E}, \mathcal{X}] \longrightarrow \mathcal{X}$$

preserves finite limits where  $\text{ev}_A : [\mathcal{A}, \mathcal{X}] \rightarrow \mathcal{X}$  is evaluation at  $A \in \mathcal{A}$ . By Lemma 7.1,

$$\text{ev}_A \circ \text{Lan}_J \cong \bigoplus_{U \leq A} \text{ev}_U .$$

Each evaluation  $\text{ev}_U$  preserves limits so their direct sum does.  $\square$

**Theorem 7.7.** *For the setting of Section 3, the adjunction (5.14) is an equivalence. Indeed, the category  $\text{Agp}$  of abelian groups can be replaced by any additive category  $\mathcal{X}$  in which idempotents split and finite direct sums exist, yielding an equivalence as announced at (1.1).*

## 8. EXAMPLES OF THEOREM 7.7

**Example 8.1.** We begin with a baby version of the Dold-Puppe-Kan Theorem. Let  $\text{Pt}\mathcal{X}$  denote the category whose objects are split epimorphisms in  $\mathcal{X}$ , the morphisms are morphisms of the epimorphisms which commute with the splittings; this is what Bourn [6] calls the category of points in  $\mathcal{X}$ . Take  $\mathbb{P}$  to be the free-living adjunction  $\mu^* \dashv \mu$  with identity counit  $\mu^* \circ \mu = 1$ . So  $[\mathcal{P}, \mathcal{X}] \cong \text{Pt}\mathcal{X}$ . Let  $\mathcal{M}$  consist of all the monomorphisms. Then  $\mathcal{R}$  contains only the identities. Theorem ?? yields

$$\text{Pt}\mathcal{X} \simeq \mathcal{X} \times \mathcal{X} .$$

This example also shows the necessity of  $\mathcal{X}$  having homs enriched in abelian groups (not merely commutative monoids). We need  $\mathcal{X}$  to have kernels of split epimorphisms and coproducts already. If we also ask that it have finite products then considering the split epimorphism  $X \times Y \rightarrow Y$  given by the projection, the counit is the canonical map  $X + Y \rightarrow X \times Y$ , so if this is invertible we have hom enrichment in commutative monoids. Now considering the codiagonal  $X + X \rightarrow X$ , split by one of the injections, it is not hard to show that  $1_X$  has an additive inverse.

**Example 8.2.** ([12, 13, 22]) Applying Theorem 7.7 to Example 4.2 yields that  $[\Delta_{\perp, \top}, \mathcal{X}]$  is equivalent to the category of chain complexes in  $\mathcal{X}$ .

**Example 8.3.** Applying Theorem 7.7 to Example 4.3 yields  $[\mathbb{I}, \mathcal{X}] \simeq [\Delta_{\text{inj}}, \mathcal{X}]$ , the category of semi-simplicial objects in  $\mathcal{X}$ .

Here are some examples of the general type described in Example 4.1.

**Example 8.4.** A *(set) species* in the sense of Joyal [19] is a functor  $F : \mathfrak{S} \longrightarrow \text{Set}$  where  $\mathfrak{S}$  is the category of sets and bijective functions. A *pointed-set species* is a functor  $F : \mathfrak{S} \longrightarrow 1/\text{Set}$ . An *R-module species* is a functor  $F : \mathfrak{S} \longrightarrow \text{Mod}^R$ ; the case where  $R$  is a field is the basic situation of [20].

Following [10], we write  $\text{FI}$  for the category of finite sets and injective functions. We then see that  $\mathcal{M} = \text{FI}$  and  $\mathcal{E} = \mathfrak{S}$ , while  $\mathcal{P} = \text{FI}\sharp$ , the category of injective partial functions.

**Corollary 8.5.** [10] *The functor*

$$\widehat{M} : [\mathfrak{S}, \text{Mod}^R] \longrightarrow [\text{FI}_\#^R, \text{Mod}^R]$$

*is an equivalence of categories.*

**Example 8.6.** Another example relevant to [21] is the category  $\mathcal{A} = \text{FIVect}_{\mathbb{F}_q}$  of finite vector spaces over the field  $\mathbb{F}_q$  of cardinality  $q$  (a prime power) and injective linear functions. Let  $\mathcal{E} = \mathfrak{S}\mathfrak{L}_q$  be the category of finite  $\mathbb{F}_q$ -vector spaces and linear isomorphisms. Then  $\mathcal{P} = \text{FI}_\#^R$  is the category of finite  $\mathbb{F}_q$ -vector spaces and injective partial linear functions.

**Corollary 8.7.** *The functor*

$$\widehat{M} : [\mathfrak{S}\mathfrak{L}_q, \text{Mod}^R] \longrightarrow [\text{FI}_\#^R, \text{Mod}^R]$$

*is an equivalence of categories.*

**Example 8.8.** Here are a few examples of categories  $\mathcal{A}$  as in Example 4.1 to which Theorem 7.7 applies with  $\mathcal{E}$  the epimorphisms and  $\mathcal{M}$  the monomorphisms:

- (a) the category of finite abelian groups and group morphisms;
- (b) the category of finite abelian  $p$ -groups and group morphisms;
- (c) the category of finite sets and all functions.

Theorem 7.7 also applies to  $\mathcal{M}$  in place of  $\mathcal{A}$  in these examples. Then  $\mathcal{E}$  is replaced by the groupoid of invertible morphisms in  $\mathcal{A}$ . In case of example (a), the paper [17] describes the groupoid being represented in  $\mathcal{X}$ .

**Example 8.9.** Consider the “algebraic” simplicial category  $\Delta_+$  whose objects are all the natural numbers and whose morphisms  $\xi : m \longrightarrow n$  are order-preserving functions

$$\xi : \{0, 1, \dots, m-1\} \longrightarrow \{0, 1, \dots, n-1\} .$$

Put  $\mathcal{A} = \Delta_+^{\text{op}}$ . Take  $\mathcal{M}$  in  $\mathcal{A}$  to consist of the surjections in  $\Delta_+$ . Pushouts of surjections along arbitrary morphisms exist in  $\Delta_+$ . Then  $\mathcal{E} = \Delta_{+\text{inj}}^{\text{op}}$  and  $\mathcal{P}$  is the opposite of the category whose morphisms  $m \longrightarrow n$  are cospans

$$m \xrightarrow{\xi} r \xleftarrow{\sigma} n$$

in  $\Delta_+$  with  $\sigma$  surjective. We could also take the “topological” simplicial category  $\Delta$  (omit the object 0) to obtain a reinterpretation of the preoperads in  $\mathcal{X}$  in the sense of [3].

**Example 8.10.** Here is a rather trivial example involving  $\Delta$ . Take  $\mathcal{A}$  to be the category of non-empty ordinals and morphisms the order-preserving functions which preserve first element. Let  $\mathcal{M}$  be the class of morphisms which are inclusions of initial segments. This is part of a factorization system where  $\mathcal{E} = \Delta_{\perp \neq \top}$  is the category of ordinals with distinct first and last element and morphisms the order-preserving functions which preserve first and last element. Sometimes  $\mathcal{E}$  is called the category of *intervals*; there is a duality isomorphism

$$\mathcal{E} \cong \Delta^{\text{op}} .$$

In this case, not only do we have the equivalence

$$[\mathcal{E}, \mathcal{X}] \simeq [\mathcal{P}, \mathcal{X}]$$

of Theorem 7.7, we actually also have an isomorphism

$$\mathcal{P} \cong \mathcal{E}.$$

**Example 8.11.** Take  $\mathcal{A}$  to be a (partially) ordered set with finite infima and the descending chain condition. Then every morphism is a monomorphism and the strong epimorphisms are equalities. So  $\mathcal{E}$  is the discrete category  $\text{ob}\mathcal{A}$  on the set of elements of the ordered set. The reader may like to contemplate the case where  $\mathcal{A}$  is the set of strictly positive integers ordered by division.

## 9. REMARKS ON IDEMPOTENTS

Define a relation on any monoid  $M$  by  $a \sqsubseteq b$  when  $ba = a$ . This relation is transitive; indeed, we have a stronger property.

**Proposition 9.1.** *If  $ub \sqsubseteq a$  and  $vc \sqsubseteq b$  then  $uvc \sqsubseteq a$ .*

*If  $u_i a_i \sqsubseteq a_{i-1}$  for  $1 \leq i \leq n$  then  $u_1 \dots u_n a_n \sqsubseteq a_0$ .*

*Proof.* For the first sentence, the assumptions are  $aub = ub$  and  $bvc = vc$ . So  $auvc = aubvc = ubvc = uvc$ ; that is,  $uvc \sqsubseteq a$ . The second sentence follows by induction.  $\square$

The relation is only reflexive for idempotents: clearly  $a \sqsubseteq a$  is equivalent to  $aa = a$ .

The unit 1 of the monoid is a largest element in the sense that  $a \sqsubseteq 1$  for all  $a \in M$ . That is, 1 is the empty meet. However, not all meets need exist. A *meet* for  $a, b \in M$  is an element  $a \wedge b$  with  $a \wedge b \sqsubseteq a$  and  $a \wedge b \sqsubseteq b$ , and, if  $x \sqsubseteq a$  and  $x \sqsubseteq b$  then  $x \sqsubseteq a \wedge b$ . In particular,  $a \wedge b$  must be an idempotent. Meets of lists of  $n$  elements are defined in the obvious way and, for  $n \geq 2$ , can be constructed from iterated binary meets when they exist.

**Proposition 9.2.** *If  $ab \sqsubseteq b$  and  $a \sqsubseteq a$  then  $a \wedge b = ab$ .*

*If  $a_1, \dots, a_n$  are idempotents such that  $a_i a_j \sqsubseteq a_j$  for  $i \leq j$  then  $a_1 \wedge \dots \wedge a_n = a_1 \dots a_n$ .*

*Proof.* For the first sentence, we are told that  $ab \sqsubseteq b$ , while  $a \sqsubseteq a$  implies  $aa = a$ , and so  $aaab = ab$ , yielding  $ab \sqsubseteq a$ . For the second sentence the result is clear for  $n = 1$  since we suppose  $a_1$  idempotent. Assume the result for  $n - 1$ ; so  $a_1 \wedge \dots \wedge a_{n-1} = a_1 \dots a_{n-1}$ . Apply the second sentence of Proposition 9.1 to the inequalities  $a_i a_n \sqsubseteq a_n$  to deduce  $a_1 \dots a_{n-1} a_n \sqsubseteq a_n$ . So, by the first sentence,  $a_1 \dots a_n = a_1 \dots a_{n-1} \wedge a_n = a_1 \wedge \dots \wedge a_{n-1} \wedge a_n$ , as required.  $\square$

Notice that, if  $ab = ba$  and  $b$  is idempotent, then  $bab = abb = ab$ , so  $ab \sqsubseteq b$ . So the proposition applies to commuting idempotents.

Now suppose we have a ring  $R$ . We can apply our results to the multiplicative monoid of  $R$ . We say idempotents  $e$  and  $f$  in  $R$  are *orthogonal* when  $ef = fe = 0$ . A list  $e_0, e_1, \dots, e_n$  of idempotents is *orthogonal* when each pair in the list is orthogonal. The list is *complete* when  $e_0 + e_1 + \dots + e_n = 1$ . An easy induction shows that a complete list of idempotents is orthogonal if and only if  $e_i e_j = 0$  for  $i \leq j$ .

For each  $a \in R$ , put  $\bar{a} = 1 - a$ . Clearly if  $a$  is idempotent, so is  $\bar{a}$ .

Let  $R^\circ$  denote the ring obtained from  $R$  by reversing multiplication.

**Proposition 9.3.** (a)  *$a \sqsubseteq b$  in  $R$  if and only if  $\bar{b} \sqsubseteq \bar{a}$  in  $R^\circ$ .*

(b) *For  $b$  an idempotent,  $ab \sqsubseteq b$  in  $R$  if and only if  $\bar{a}\bar{b} \sqsubseteq \bar{b}$  in  $R^\circ$ .*

(c) If  $a$  and  $b$  are idempotents and  $ab \sqsubseteq b$  in  $R$  then  $e_0 = ab, e_1 = \bar{a}b, e_2 = \bar{b}$  is a complete list of orthogonal idempotents.

*Proof.* (a)  $\bar{b} \sqsubseteq \bar{a}$  in  $R^\circ$  means  $(1-b)(1-a) = 1-b$  in  $R$ ; that is,  $ba = a$  which means  $a \sqsubseteq b$  in  $R$ .

(b)  $\bar{a}\bar{b} \sqsubseteq \bar{b}$  in  $R^\circ$  means  $\bar{b}\bar{a}\bar{b} = \bar{b}\bar{a}$  in  $R$ . That is,  $(1-b)(1-a)(1-b) = (1-b)(1-a)$ . That is,  $1-a-b+ab-b+ba+bb-bab = 1-b-a+ba$ . That is,  $bab = ab$ , which is  $ab \sqsubseteq b$  in  $R$ .

(c) We already know  $e_0$  and  $e_2$  are idempotent. They are also orthogonal:  $e_0e_2 = ab(1-b) = ab - ab = 0$  and  $e_2e_0 = (1-b)ab = ab - bab = 0$ . Therefore  $e_0 + e_2 = ab + 1 - b = 1 - (1-a)b = \bar{e}_1$  is idempotent. So  $e_1$  is idempotent and  $e_0 + e_1 + e_2 = 1$ . The calculations  $e_0e_1 = ab(1-a)b = ab - abab = 0$  and  $e_1e_2 = (1-a)b(1-b) = b - ab - b + abb = 0$  complete the proof.  $\square$

We can extend part (c) inductively to obtain:

**Proposition 9.4.** *Suppose  $a_1, \dots, a_n$  are idempotents such that  $a_i a_j \sqsubseteq a_j$  for  $i \leq j$ . Then  $e_i = \bar{a}_i a_{i+1} a_{i+2} \dots a_n$  for  $0 \leq i \leq n$  (in particular,  $e_0 = a_1 a_2 \dots a_n$  and  $e_n = \bar{a}_n$ ) defines a complete list of orthogonal idempotents.*

Suppose  $\mathcal{X}$  is an additive category in which idempotents split. Our results apply to the endomorphism monoid  $\mathcal{X}(A, A)$  of each object  $A \in \mathcal{X}$ . If  $a$  is an idempotent on  $A$ , we have a splitting:

$$\begin{array}{ccc} A & \xrightarrow{a} & A \\ r_a \downarrow & \nearrow i_a & \downarrow r_a \\ aA & \xrightarrow{1_{aA}} & aA \end{array}$$

Yet, we also have a splitting for  $\bar{a} = 1 - a$  which incidentally provides a kernel  $\bar{a}A$  for  $a$  and so a direct sum decomposition of  $A$ :

$$A \cong \bar{a}A \oplus aA.$$

More generally, for any complete list  $e_0, e_1, \dots, e_n$  of orthogonal idempotents in  $\mathcal{X}(A, A)$ , we obtain a direct sum decomposition

$$A \cong e_0 A \oplus e_1 A \oplus \dots \oplus e_n A.$$

## 10. WHEN $\mathcal{X}$ IS SEMIABELIAN

Semiabelian categories include the category  $\text{Grp}$  of (not necessarily abelian) groups and group morphisms. In [7] Dominique Bourn gave a version of the Dold-Puppe-Kan Theorem (Example 8.2) for the case where the codomain category  $\mathcal{X}$  was semiabelian. In that case it asserted monadicity of the right adjoint in

$$[\mathcal{D}, \mathcal{X}]_{\text{pt}} \xrightleftharpoons[\widetilde{M}]{\widehat{M}} [\mathcal{P}, \mathcal{X}]. \quad (10.32)$$

In this section, we provide a version of this for  $\mathcal{P}$  as in Section 3 in place of  $\Delta^{\text{op}}$ . However, we require an extra assumption related to idempotents.

**Assumption 10.1.** The maximal proper elements of  $\text{Sub}A$  can be listed  $m_1, \dots, m_n$  such that the idempotents  $c_i = m_i \circ m_i^*$  on  $A$  satisfy  $c_j \circ c_i \circ c_j = c_j \circ c_i$  for all  $i < j$ .

Throughout we assume our category  $\mathcal{X}$  has zero morphisms (that is, has homs enriched in pointed sets).

We begin by providing a non-additive version of the material at the end of Section 9 on idempotents.

**Proposition 10.2.** *Suppose the category  $\mathcal{X}$  has kernels of idempotents. Let  $e, f$  be idempotents on an object  $A$  of  $\mathcal{X}$ . If  $e \circ f \circ e = e \circ f$  then the intersection of the kernels of  $e$  and  $f$  exists.*

*Proof.* Let  $k : K \rightarrow A$  be the kernel of  $e$ . Since  $e \circ f \circ k = e \circ f \circ e \circ f = 0$ , there exists a unique  $g$  with  $f \circ k = k \circ g$ . Then  $k \circ g \circ g = f \circ k \circ g = f \circ f \circ k = f \circ k = k \circ g$  and  $k$  is a monomorphism. So  $g$  is idempotent. Then the kernel  $\ell : L \rightarrow K$  of  $g$  is easily verified to be the intersection of the kernels of  $e$  and  $f$ .  $\square$

Protomodular categories were defined by Bourn [6]: a category  $\mathcal{X}$  (with zero morphisms) is *protomodular* when it is finitely complete and, for each object  $A$ , the functor  $\ker : \text{Pt}A \rightarrow \mathcal{X}$  is conservative. Here  $\text{Pt}A$  is the category whose objects  $(p, X, s)$  consist of morphisms  $p : X \rightarrow A, s : A \rightarrow X$  with  $p \circ s = 1_A$ , and whose morphisms  $f : (p, X, s) \rightarrow (q, Y, t)$  are morphisms  $f : X \rightarrow Y$  such that  $q \circ f = p$  and  $f \circ s = t$ . Also the functor  $\ker$  takes  $(p, X, s)$  to the kernel of  $p$ .

The following property is sometimes [5] taken as the definition of protomodular.

**Lemma 10.3.** *In a protomodular category, if  $(p, X, s)$  is an object of  $\text{Pt}A$  and  $k : K \rightarrow X$  is the kernel of  $p$  then  $s : A \rightarrow X, k : K \rightarrow X$  are jointly strongly epimorphic.*

*Proof.* Suppose  $m : Y \rightarrow X$  is a monomorphism and  $m \circ u = s, m \circ v = k$  for some  $u, v$ . Then  $m : (p \circ m, Y, u) \rightarrow (p, X, s)$  is a morphism of  $\text{Pt}A$ . Using  $v$ , we see that  $m$  induces an isomorphism between the kernel of  $p \circ m$  and  $K$ . Since  $\ker : \text{Pt}A \rightarrow \mathcal{X}$  is conservative,  $m : (p \circ m, Y, u) \rightarrow (p, X, s)$  is invertible. So  $m$  is invertible.  $\square$

**Proposition 10.4.** *Let  $a_1, \dots, a_n$  be a list of idempotents on an object  $A$  of a protomodular category  $\mathcal{X}$ . Suppose  $a_i \circ a_j \circ a_i = a_i \circ a_j$  for  $i < j$ . Suppose  $a_i = m_i \circ m_i^*$  is a splitting of  $a_i$  via a subobject  $m_i : A_i \rightarrow A$  and retraction  $m_i^* : A \rightarrow A_i$ . Let  $k_i : K_i \rightarrow A$  be the kernel of  $m_i^*$  (or equally of  $a_i$ ). Then the morphisms  $m_1, \dots, m_n$  along with the inclusion  $\cap_i \ker m_i^* \rightarrow A$  are jointly strongly epimorphic.*

*Proof.* By Lemma 10.3, for each  $i$ , the morphisms  $m_i : A_i \rightarrow A$  and  $k_i : K_i \rightarrow A$  are jointly strongly epimorphic; we will loosely say “ $A_i$  and  $K_i$  cover  $A$ ”.

If  $i > 1$  then  $a_1 \circ a_i \circ k_1 = a_1 \circ a_i \circ a_1 \circ k_1 = 0$ , and so  $a_i \circ k_1$  lands in  $K_1$ , providing a factorization  $a_i \circ k_1 = k_1 \circ a_i^1$ . Now  $a_i^1$  is also an idempotent, and, for  $1 < i < j$ ,  $k_1 \circ a_i^1 \circ a_j^1 \circ a_i^1 = a_i \circ a_j \circ a_i \circ k_1 = a_i \circ a_j \circ k_1 = k_1 \circ a_i^1 \circ a_j^1$ , and so  $a_i^1 \circ a_j^1 \circ a_i^1 = a_i^1 \circ a_j^1$ . Clearly the splitting  $A_i^1$  of  $a_i^1$  is contained in  $A_i$ .

The kernel  $K_i^1$  of  $a_i^1$  is  $K_i^1 = K_1 \cap K_i$  since  $a_1 \circ x = a_i \circ x = 0$  is equivalent to  $x = k_1 \circ y$  and  $k_1 \circ a_i^1 \circ y = a_i \circ k_1 \circ y = s_i \circ x = 0$ ; so in fact  $a_i^1 \circ y = 0$ .

We know that  $A$  may be covered by  $A_1$  and  $K_1$ . By Lemma 10.3 again, we know, for each  $i > 1$ , that  $K_1$  may be covered by the splitting of  $A_i^1$  and the kernel  $K_i^1 = K_1 \cap K_i$  of  $a_i^1$ . Since  $A_i^1 \leq A_i$ , we see that  $A$  may be covered by  $A_1, A_i$ , and  $K_1 \cap K_i$ .

Now continue inductively.  $\square$

**Theorem 10.5.** *Suppose  $\mathcal{P}$  is as in Section 3 and  $\mathcal{X}$  is protomodular (with zero morphisms) with finite coproducts. Then the components of the adjunction (10.32) are strong epimorphisms.*

*Proof.* The counit has components

$$\sum_{B \preceq_m A} \bigcap_{C \prec_n B} \ker Tn^* \longrightarrow TA.$$

We prove this is a strong epimorphism by induction on the number  $k$  of maximal proper subobjects  $A_1, \dots, A_k$  of  $A$ , with  $m_i : A_i \rightarrow A$ . The result is clear for  $k = 1$ . For  $k > 1$ , consider the following diagram.

$$\begin{array}{ccc} \sum_i \sum_{B \preceq_m A} \bigcap_{C \prec_n B} \ker Tn^* & \longrightarrow & \sum_i TA_i \\ \delta \downarrow & & \downarrow \gamma \\ \sum_{B \prec_m A} \bigcap_{C \prec_n B} \ker Tn^* & \xrightarrow{\alpha} & FA \xleftarrow{\beta} \bigcap_{C \prec_n A} \ker Tn^* \end{array}$$

The component of the counit is strongly epimorphic if and only if  $\alpha$  and  $\beta$  are jointly strongly epimorphic. The top row is a coproduct of components of the counit already known to be strongly epimorphic by induction. So it suffices to show that  $\gamma$  and  $\beta$  are jointly strongly epimorphic. Rewriting the domain of  $\beta$  as

$$\bigcap_{C \prec_n A} \ker Tn^* = \bigcap_{i=1}^k \ker Tm_i^*,$$

we see that Proposition 10.4 applies to yield what we want.  $\square$

A category  $\mathcal{X}$  is *semiabelian* [18] when it has zero morphisms, is protomodular, is Barr exact, and has finite coproducts. A category is *regular* [1] when it is finitely complete, and has the (strong epimorphism, monomorphism)-factorization system existing and stable under pullbacks. It follows that every strong epimorphism is regular (that is, a coequalizer); see [9] for a proof. A category is *Barr exact* when it is regular and every equivalence relation is a kernel pair.

We mentioned Bourn's category  $\text{Pt}\mathcal{X}$  in Example 8.1. We will use the following routine fact.

**Lemma 10.6.** *If  $\mathcal{X}$  is a semiabelian category then the functor  $\text{Pt}\mathcal{X} \rightarrow \mathcal{X}$ , sending each split epimorphism to its kernel, preserves strong epimorphisms.*

*Proof.* A strong epimorphism

$$\begin{array}{ccc} X & \xrightarrow{g} & Y \\ s \uparrow & & \uparrow t \\ A & \xrightarrow{f} & B \\ & p \downarrow & \downarrow q \end{array}$$

in  $\text{Pt}\mathcal{X}$  has  $f$  and  $g$  strong epimorphisms in  $\mathcal{X}$ . From this it is easily verified that the square involving the downward-pointing arrows is a pushout. Factor the morphism in

$\text{Pt}\mathcal{X}$  as

$$\begin{array}{ccccc}
 X & \xrightarrow{g_1} & Z & \xrightarrow{f_1} & Y \\
 \uparrow s & & \uparrow t_1 & & \uparrow t \\
 \downarrow p & & \downarrow q_1 & & \downarrow q \\
 A & \xrightarrow{1_A} & A & \xrightarrow{f} & B
 \end{array}$$

in which the right-hand square involving the downward-pointing arrows is a pullback. The induced morphism  $\ker q_1 \rightarrow \ker q$  is invertible. Semiabelian categories are Maltsev [18]. Therefore, as a comparison morphism to the pullback in a pushout square in a Maltsev exact category,  $g_1$  is a strong epimorphism (see Theorem 5.7 of [8]). The induced morphism  $\ker p \rightarrow \ker q_1$  is the pullback of  $g_1$  along  $\ker q_1 \rightarrow Z$  and so is a strong epimorphism by regularity.  $\square$

**Theorem 10.7.** *If  $\mathcal{P}$  is as in Section 3 and  $\mathcal{X}$  is semiabelian then the adjunction of Theorem ?? is (crudely) monadic.*

*Proof.* By Theorem 10.5, the right adjoint (tilde) is conservative (since this is logically equivalent to the counit being a strong epimorphism). Since  $\mathcal{X}$  is semiabelian, it has coequalizers. Therefore  $[\mathcal{P}, \mathcal{X}]$  has coequalizers, so, for crude monadicity [28], it suffices to show that tilde preserves coequalizers of reflexive pairs.

Both  $[\mathcal{P}, \mathcal{X}]$  and  $[\mathcal{P}, \mathcal{X}]_{\text{pt}}$  are semiabelian. A limit-preserving functor between semiabelian categories preserves coequalizers of reflexive pairs provided it preserves strong (= regular) epimorphisms (see Lemma 5.1.12 of [4]).

Let  $q : S \rightarrow T$  be a strong epimorphism in  $[\mathcal{P}, \mathcal{X}]$ . Each  $q_A : SA \rightarrow TA$  is a strong epimorphism. We must show each  $\tilde{q}_A : \tilde{S}A \rightarrow \tilde{T}A$  is a strong epimorphism.

Recall that  $\tilde{T}A$  is calculated by a sequence of kernels of split epimorphisms. This sequence depends only on the object  $A$  and the category  $\mathcal{P}$ , not on the particular functor  $T$ . The desired result follows on repeated application of Lemma 10.6.  $\square$

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